Software Test Plan

Project: File Compression Tool (Basic ZIP Implementation)

Version: 1.0

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1. Introduction

Purpose: This document defines the test plan for the File Compression Tool v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing the tool prior to submission.

Scope: Testing covers the tool's core CLI features — compressing files/folders into a ZIP archive, extracting an archive, listing archive contents, validating archive structure, verifying entry integrity, and rejecting unsafe (path-traversing) extraction targets. GUI design, cloud storage integration, multi-volume archives, and encryption are out of scope, consistent with the SRS.

References: File Compression Tool SRS v1.0; Software Architecture and Design Specification v1.0; OWASP path-traversal (zip-slip) guidance.

Definitions: ZIP, CLI, CRC, SHA-256, SRS, SAD, RTM, path traversal.

2. Test Items

- CLI Interface / Command Orchestrator
- Archive Writer (compression)
- Archive Parser (listing & validation)
- Extraction Module
- Path/Security Validator
- Integrity Module (CRC / SHA-256)
- Resource Manager (cleanup)
- Logging Module

3. Features to be Tested

Features mapped to SRS requirement IDs:

Req ID	Feature
FCT-F-001	Select compression inputs (files/folders)
FCT-F-002	Create ZIP archive
FCT-F-003	Preserve directory structure in archive
FCT-F-004	Compress file data
FCT-F-005	Report compression status/summary
FCT-F-006	Select extraction destination

Req ID	Feature
FCT-F-007	Extract archive entries
FCT-F-008	Block path traversal on extraction
FCT-F-009	Reject malformed/corrupted archive
FCT-F-010	List archive contents without extracting
FCT-F-011	Apply duplicate-file overwrite/skip/rename policy
FCT-F-012	Verify entry integrity after extraction
FCT-F-013	Preserve binary file contents exactly
FCT-F-014	Clean up temporary resources
FCT-F-015	Display operation completion summary
FCT-NF-001	Performance: ≤ 2s per 100 MB
FCT-NF-002	Memory: ≤ 512 MB peak for 1 GB workload
FCT-NF-003	Clear, actionable error messages
FCT-NF-004	Round-trip content correctness
FCT-NF-005	Cross-platform path handling (Windows/Linux/macOS)
FCT-SR-001	Path traversal protection
FCT-SR-002	ZIP structure validation
FCT-SR-003	Entry integrity verification
FCT-SR-004	No sensitive data (file contents) in logs
FCT-SR-005	Non-destructive overwrite policy

4. Features Not to be Tested

- Correctness of Python's standard library internals (zipfile, hashlib) — assumed tested by CPython
- Underlying OS file system reliability — assumed provided by the operating system
- GUI — not built in this basic implementation
- Password-protected / encrypted archives — out of scope per SRS
- Multi-volume archive support — out of scope per SRS

5. Test Approach / Strategy

Levels:

- Unit tests — module level (Archive Writer, Archive Parser, Extraction Module, Path/Security Validator, Integrity Module), using pytest
- Integration tests — CLI Interface → Command Orchestrator → core module chain
- System tests — end-to-end compress → extract round trip via the CLI
- Acceptance tests — team walkthrough of all SRS-mapped features before submission

Types:

- Functional testing (core compress/extract/list features)
- Regression testing (re-run the pytest suite after every change)
- Performance testing (timing runs against FCT-NF-001, FCT-NF-002)

- Security testing (path traversal, malformed archives)
- Usability testing (manual review of CLI output and error messages)

Entry Criteria: Stable build on the main branch; sample test files/folders and crafted test archives prepared; pytest environment set up.

Exit Criteria: All planned test cases executed; 0 critical defects open; all SRS-mapped requirements covered by at least one test case.

5.1 Security Validation

- Craft a zip-slip archive (entry paths like ../../etc/passwd) and confirm extraction is rejected (FCT-SR-001)
- Feed a corrupted/truncated ZIP file and confirm the tool reports an error instead of crashing (FCT-F-009, FCT-SR-002)
- Inspect log output after several operations to confirm no file contents appear, only metadata (FCT-SR-004)
- Pass invalid/unexpected CLI arguments (empty input, non-existent paths, wrong flags) and confirm graceful error handling
- Extract an archive with a duplicate filename already present at the destination and confirm the overwrite/skip/rename policy prevents silent data loss (FCT-SR-005)

6. Test Environment

Hardware: Standard development laptops (Windows, Linux, and/or macOS, to validate FCT-NF-005).

Software: Python 3.x, pytest, the project's CLI source.

Tools: pytest (unit/integration tests), Python's time and tracemalloc modules for basic performance/memory checks, manual terminal testing for CLI usability, Git for version control.

Test Data: Sample small text files, a large file (≥100 MB) for performance testing, nested folders (to test directory structure preservation), files with duplicate names, a hand-crafted malicious archive with path-traversal entries, and a deliberately corrupted ZIP file.

7. Test Schedule

Milestones:

- Test case design: 16-09-2026
- Test environment & sample data setup: After implementation build is available
- Test execution start: To be scheduled
- Test execution end: To be scheduled
- Acceptance walkthrough / sign-off: To be scheduled

8. Test Deliverables

- Test Plan (this document)
- Test Cases (pytest scripts and manual CLI checklists)
- Test Data (sample and crafted archives/files)
- Test Execution Logs
- Defect Reports

- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Aakash Agarwal	Prepare the test plan, coordinate execution across the team
Test Engineer	Abhijna Marathe	Design and execute test cases, log defects
Developer	Adarsh Prakash	Support defect fixes and triage
Product Owner	Abhirup Roy	Approve test results, sign off on submission readiness

Note: with a 4-person team, roles are a primary point of contact rather than an exclusive assignment — in practice everyone writes and runs tests for the module(s) they built.

10. Risks and Mitigation

Risk	Mitigation
Large test files slow down local test runs	Keep the performance-test file set small enough for a laptop (e.g. ~100–200 MB) while still exercising the NF-001/NF-002 targets
Team members test on different OSes with inconsistent results	Run the full pytest suite on at least one Windows and one Unix-like machine before sign-off (FCT-NF-005)
Crafted malicious/corrupted test archives accidentally get lost or overwritten	Keep test data in a dedicated, version-controlled test-fixtures folder
Limited time before submission deadline	Prioritize security and round-trip correctness tests first, since those map to the most heavily weighted SRS requirements

11. Assumptions & Dependencies

- Python 3.x and pytest are available on all team members' test machines
- Sample and crafted test archives will be prepared before execution begins
- No external dependencies beyond the Python standard library are introduced during development

12. Suspension & Resumption Criteria

Suspend testing if:

- The Python/pytest environment is broken for more than 2 hours
- The current build blocks more than 30% of planned test cases

Resume testing if:

- Blocking defects are resolved
- The environment is stable and the full pytest suite can run

13. Test Case Management & Traceability

An RTM maps SRS requirements to test cases. Example mappings:

Req ID	Test Case(s)
FCT-F-002 (Create ZIP archive)	TC-Comp-01, TC-Comp-02

Req ID	Test Case(s)
FCT-F-007 (Extract archive entries)	TC-Extr-01, TC-Extr-02
FCT-SR-001 (Path traversal protection)	TC-Sec-01
FCT-SR-002 (ZIP structure validation)	TC-Sec-02
FCT-NF-001 (Performance)	TC-Perf-01
FCT-NF-004 (Round-trip correctness)	TC-Corr-01

14. Test Metrics & Reporting

Metrics collected:

- % of test cases executed
- % passed / failed
- Defect count and severity
- SRS requirement coverage (% of requirement IDs with a passing test case)

Reports:

- Execution status updates during the test window
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Product Owner		